

Evaluation of 6 Models for BigCodeBench/660

Task Overview

Problem Description: Scale the "x" and "y" arrays using the standard scaler of sklearn and plot them with given labels. Each pair of x and y arrays are scaled independently and plotted as a separate series with a label. The function should output a matplotlib.figure.Figure object.

Model 1 Evaluation

model1_efficiency

Total steps: 12

Step Analysis:

- <step 1> Problem understanding - **Essential**
- <step 2> Import requirements - **Essential**
- <step 3> Function signature understanding - **Essential**
- <step 4> Create figure/axes - **Essential**
- <step 5> Iteration approach - **Essential**
- <step 6> Reshaping explanation - **Essential**
- <step 7> Independent scaling approach - **Essential**
- <step 8> fit_transform usage - **Essential**
- <step 9> Output reshaping discussion - **Non-essential** (overly detailed about reshape options)
- <step 10> Plotting approach - **Essential**
- <step 11> Legend addition - **Essential**
- <step 12> Return statement - **Essential**

Efficiency Score: 11/12 essential steps. Model is verbose but thorough, with one step providing unnecessary detail about reshaping options.

model1_logic

Step Coherence:

- All steps logically follow from previous steps
- Clear progression from problem understanding → imports → function design → implementation details
- No logical gaps identified

Internal Consistency:

- Maintains consistent approach throughout
- Correctly identifies need for independent scalers
- No contradictions found

model1_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- Correct understanding of StandardScaler requiring 2D input
- Correct approach to independent scaling
- Accurate matplotlib usage

Edge Case Handling:

- No consideration of empty arrays
- No validation of input lengths matching
- No error handling mentioned

Model 2 Evaluation

model2_efficiency

Total steps: 5

Step Analysis:

- <step 1> Parameter understanding - **Essential**
- <step 2> Scaling requirements with reshaping - **Essential**
- <step 3> Plotting and legend approach - **Essential**
- <step 4> Return requirement - **Essential**
- <step 5> Function wrapping mention - **Non-essential** (redundant statement)

Efficiency Score: 4/5 essential steps. Very concise but covers main points.

model2_logic

Step Coherence:

- Steps flow logically but are very high-level
- <step 2> coherent - follows from understanding inputs
- <step 3> coherent - follows from scaling
- <step 4> coherent
- <step 5> somewhat redundant but coherent

Internal Consistency:

- No contradictions
- Maintains focus on independent scaling

model2_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- Correctly identifies 2D input requirement
- Accurate understanding of StandardScaler

Edge Case Handling:

- No edge cases considered
- No error handling discussed

Model 3 Evaluation

model3_efficiency

Total steps: 9

Step Analysis:

- <step 1> Problem overview and imports - **Essential**
- <step 2> Figure/axes creation - **Essential**
- <step 3> Iteration approach - **Essential**
- <step 4> Accessing elements - **Non-essential** (redundant explanation)
- <step 5> Reshaping explanation - **Essential**
- <step 6> Independent scaling with new instances - **Essential**
- <step 7> Plotting approach - **Essential**
- <step 8> Legend addition - **Essential**
- <step 9> Return statement - **Essential**

Efficiency Score: 8/9 essential steps. Well-structured with one redundant step.

model3_logic

Step Coherence:

- All steps coherent and build logically
- Clear progression through implementation
- <step 4> is somewhat redundant after <step 3> but still coherent

Internal Consistency:

- Consistently emphasizes independent scaling
- No contradictions
- Maintains same approach throughout

model3_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- Correct understanding of all components
- Accurate sklearn and matplotlib usage

Edge Case Handling:

- Mentions assumption about equal lengths but doesn't implement validation
 - No error handling
-

Model 4 Evaluation

model4_efficiency

Total steps: 9 (includes actual code implementation)

Step Analysis:

- <step 1> Import statement - **Essential**
- <step 2> Input validation - **Essential** (good practice)
- <step 3> Figure creation - **Essential**
- <step 4> Iteration overview - **Essential**
- <step 5> Scaling process - **Essential**
- <step 6> Flatten for plotting - **Essential**
- <step 7> Plot with label - **Essential**
- <step 8> Add legend - **Essential**
- <step 9> Return figure - **Essential**

Efficiency Score: 9/9 essential steps. Most comprehensive with error handling.

model4_logic

Step Coherence:

- Excellent logical flow
- Adds validation before processing (good practice)
- Each step follows naturally
- Implementation matches reasoning perfectly

Internal Consistency:

- Completely consistent
- Maintains same approach throughout

- Code implementation matches description

model4_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- All statements factually correct
- Proper use of numpy arrays and ravel()
- Correct StandardScaler usage

Edge Case Handling:

- **Excellent:** Validates input lengths
- Checks if x[i] and y[i] have same length
- Raises appropriate errors
- Most complete error handling

Model 5 Evaluation

model5_efficiency

Total steps: 4

Step Analysis:

- <step 1> Problem statement and key requirement - **Essential**
- <step 2> Technical details of scaling - **Essential**
- <step 3> Implementation approach - **Essential**
- <step 4> Final output requirements - **Essential**

Efficiency Score: 4/4 essential steps. Very concise, all steps essential.

model5_logic

Step Coherence:

- All steps coherent but very high-level
- Logical flow from problem → technical details → implementation → output
- No gaps but lacks detail

Internal Consistency:

- Consistent throughout
- No contradictions

model5_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- Statements are correct but very brief
- Mentions reshaping requirement correctly

Edge Case Handling:

- No edge cases considered
- No error handling mentioned

Model 6 Evaluation

model6_efficiency

Total steps: 6

Step Analysis:

- <step 1> Problem understanding - **Essential**

- <step 2> Input interpretation - **Essential**
- <step 3> Process breakdown - **Essential**
- <step 4> StandardScaler explanation - **Essential**
- <step 5> Legend and return requirement - **Essential**
- <step 6> Visualization enhancement suggestion - **Non-essential** (beyond requirements)

Efficiency Score: 5/6 essential steps. Last step adds unnecessary features.

model6_logic

Step Coherence:

- Steps 1-5 flow logically
- <step 6> is coherent but goes beyond problem requirements

Internal Consistency:

- Consistent approach
- Correctly identifies need for separate scalers
- No contradictions

model6_completeness

Problem Coverage:

Problem Component	Addressed?
Scale x and y arrays	Yes
Use StandardScaler	Yes
Scale independently	Yes
Plot as separate series	Yes
Use given labels	Yes
Return Figure object	Yes

Factual Correctness:

- Correct understanding of StandardScaler
- Accurate description of independent scaling

Edge Case Handling:

- No edge cases considered
- No error handling mentioned

- Suggests extra features not required
-

Summary Ranking

Best Overall: Model 4 - Most complete with error handling and actual implementation **Most Efficient:** Model 5 - Concise with all essential steps **Most Detailed:** Model 1 - Thorough explanation of each component **Best Logic:** Model 4 - Clear logical flow with validation **Most Complete:** Model 4 - Handles edge cases and validation